

FARMER PRICE INTELLIGENCE DASHBOARD

Data 2 Capstone Project

Team Members:

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BACKGROUND

Problem

According to the World Food Programme (WFP), Ghana loses roughly \$1.9 billion annually to post-harvest and market access inefficiencies. As such, smallholder farmers regularly sell their produce to local middlemen at heavily discounted prices because they lack real-time access to urban wholesale market prices. This means they sell to the nearest buyer without knowing market prices elsewhere.

Our Solution

An interactive Power BI dashboard that shows: Real-time commodity prices across regions, Where to sell for best price, When to sell (seasonal timing), and How much money farmers can expect to make in a season.

Key Success Metrics

1. Farmer Income Impact (how much they gain)
2. Regional Arbitrage Opportunities (where to sell)
3. Seasonal Timing Advantage (when to sell)
4. Market Efficiency Score (diagnostic)
5. Smart Recommendations (what to do)

DATA SOURCES

Primary Dataset

MoFA Agricultural Commodity Prices

Link: <https://srid.mofa.gov.gh/node/18>

Contents: Retail and wholesale prices of agricultural commodities, multiple regional markets, updated Nov 7, 2025, CSV format.

Direct CSV:

https://srid.mofa.gov.gh/sites/default/files/uploaded_resources/Commodity%20prices%20_04.11.25.csv

Secondary Dataset

Historical Data (1970-2017):

<https://data.gov.gh/dataset/agricultural-production-estimates-1993-2017>

Additional Resources

Ghana Open Data Portal: <https://data.gov.gh/>

Ghana Statistical Service: <https://gssweb.statsghana.gov.gh/>

World Bank Data (Ghana): <https://data.worldbank.org/country/ghana>

TECH STACK

Data Preparation: Python + Pandas

Data Visualization: Power BI Desktop

Deployment: Power BI Web

Version Control: GitHub

FEATURES

Feature 1: FARMER INCOME IMPACT

Question: How much money am I leaving on the table?

Formula: $\text{Income_Loss} = (\text{Best_Price} - \text{Worst_Price}) \times \text{Average_Farmer_Harvest_Volume}$

Example: Maize farmer in Takoradi, 100-bag harvest.

Best market (Accra): GHS 500/bag. Worst market (Takoradi): GHS 280/bag.

Loss if selling locally: $(500 - 280) \times 100 = \text{GHS } 22,000 \text{ LOST}$

Feature 2: REGIONAL PRICE ARBITRAGE

Question: If I travel to a better market, how much do I gain?

Formula: $\text{Price_Gain_}\% = (\text{Best_Regional_Price} - \text{Home_Region_Price}) / \text{Home_Region_Price} \times 100$

Example: Farmer in Takoradi: Home price GHS 280/bag, Best nearby market (Kumasi) GHS 450/bag, Gain: 60.7%

Feature 3: MARKET EFFICIENCY SCORE

Question: How much market information is MISSING in each region?

Formula: $\text{Market_Efficiency} = 1 - (\text{Price_StDev} / \text{Price_Mean})$

Example: Takoradi maize 45% efficient (INEFFICIENT - huge price gaps) vs Accra 92% efficient (TIGHT competition). Takoradi prices swing GHS 200-450, farmers have NO idea what is fair.

SOURCES:

Loses Billions in post harvest losses:

<https://www.graphic.com.gh/news/general-news/ghana-news-ghana-loses-1-9bn-to-post-harvest-losses-annually-wfp.html>

SME Competitiveness Outlook: Big money for small business Financing the Sustainable Development Goals pdf